

Ints = 1, 17, 10, 15, 14, 43, 4, 21, 23, 24, 19, 41, 42, 9, 44

1A.

$$3(1) + 2 = 5 \% 11 = 5$$

$$3(17) + 2 = 53 \% 11 = 9$$

$$3(10) + 2 = 32 \% 11 = 10$$

$$3(15) + 2 = 47 \% 11 = 3$$

$$3(14) + 2 = 44 \% 11 = 0$$

$$3(43) + 2 = 131 \% 11 = 10$$

$$3(4) + 2 = 14 \% 11 = 3$$

$$3(21) + 2 = 65 \% 11 = 10$$

$$3(23) + 2 = 71 \% 11 = 5$$

$$3(24) + 2 = 74 \% 11 = 8$$

$$3(19) + 2 = 59 \% 11 = 4$$

$$3(41) + 2 = 125 \% 11 = 4$$

$$3(42) + 2 = 128 \% 11 = 7$$

$$3(9) + 2 = 29 \% 11 = 7$$

$$3(44) + 2 = 134 \% 11 = 2$$

POS 0:	14			
POS 1:	NULL			
POS 2:	44			
POS 3:	15 →	4		
POS 4:	19 →	41		
POS 5:	1 →	23		
POS 6:	NULL			
POS 7:	42 →			9
POS 8:	24			
POS 9:	17			
POS 10:	10 →			43 →

1B.

$$3(1) + 2 = 5 \% 11 = 5$$

$$3(17) + 2 = 53 \% 11 = 9$$

$$3(10) + 2 = 32 \% 11 = 10$$

$$3(15) + 2 = 47 \% 11 = 3$$

$$3(14) + 2 = 44 \% 11 = 0$$

$$3(43) + 2 = 131 \% 11 = 10$$

$$3(4) + 2 = 14 \% 11 = 3$$

$$3(21) + 2 = 65 \% 11 = 10$$

$$3(23) + 2 = 71 \% 11 = 5$$

9/11 = 0.818, therefore load factor of 0.75 is met. So table size is doubled to 22.

$$3(24) + 2 = 74 \% 22 = 8$$

$$3(19) + 2 = 59 \% 22 = 15$$

$$3(41) + 2 = 125 \% 22 = 15$$

$$3(42) + 2 = 128 \% 22 = 18$$

$$3(9) + 2 = 29 \% 22 = 7$$

$$3(44) + 2 = 134 \% 22 = 2$$

POS 0:	14		
POS 1:			
POS 2:	44		
POS 3:	15	4	
POS 4:			
POS 5:	1	23	
POS 6:			
POS 7:	9		
POS 8:	24		
POS 9:	17		
POS 10:	10	43	21
POS 11:			
POS 12:			
POS 13:			
POS 14:			

POS 15:	19	41	
POS 16:			
POS 17:			
POS 18:	42		
POS 19:			
POS 20:			
POS 21:			

2A. $A = 7, B = 2, f(n) = 1$

$$n^{\log_b(a)} = n^{\log_2(7)}$$

$n^{\log_b(a)} = O(n^{2.80})$ is larger than $f(n)$ and for some $\epsilon < \log_2(7)$, so case 1 of the Masters Theorem applies here and proves how $T(n) \in O(n^{\log_2(7)})$

2B.

$$\text{Base Case: } T(1) = \frac{1}{6} (7(1)^{\log_2(7)} - 1) = \frac{1}{6} (7 - 1) = 6/6 = 1$$

- Therefore base case is proven

Inductive Hypothesis: Now that $T(k)$ is proved, $T(k+1)$ is going to be the inductive step. Next prove the statement that $T(k+1) = \frac{1}{6} (7^{(k+1)\log_2(7)} - 1)$ is true.

Inductive Step:

$$\text{LHS: } T(k+1) = 7T((k+1)/2) + 1$$

$$T(k+1) = \frac{1}{6} (7^{(k+1)\log_2(7)} - 1)$$

$$7(\frac{1}{6} (7^{(k+1)\log_2(7)} - 1)/2) + 1$$

$$7/6 * (7^{((k+1)/2)\log_2(7)}) + 1 - 7/6$$

- Since $2^{-\log_2(7)} = 1/7$, expression can be further simplified as 7 can be eliminated with it.

$$7/6 * ((k+1)^{\log_2(7)}) - \frac{1}{6}$$

$$\frac{1}{6} * (7 * (n+1)^{\log_2(7)} - 1)$$

- $T(n) = \frac{1}{6} (7n^{\log_2(7)} - 1)$ is therefore proven as $n = n + 1$ in the above case.

- Therefore statement is proven by induction.

3A.

$C = 1$, so doesn't really factor in.

Level 1 of Tree: 1 Node of $n = n$

Level 2 of Tree: 2 Nodes of $n/2 = n$

Level 3 of Tree: 4 Node of $n/4 = 4n$

- Stop subdividing when there are two elements left.

$n/2^k < 2$, implying $n = 2^{(k+1)}$ or $k = \log_2(n) - 1$

For $n \geq 2$, $T(n) = 4T(n/2) + n$

$T(n) = [n + 4n/2 + 4^2n/2^2 + 4^kn/2^k]$

$n[1 + 2 + 2^2 + \dots + (2)^k]$

$n[((2)^{(k+1)} - 1) / 2 - 1]$

$n[((2)^{(k+1)} - 1)]$

$T(n) = n(2^{k+1}) - 1$

- Plug value for k back in $(\log_2(n) - 1)$

$T(n) = n^2 - n$

Asymptomatic upper bound for the relation is $O(n^2)$ as it is the highest power of n in final answer.

3B.

$C = 1$, so doesn't really factor in.

Level 1 of Tree: 1 Node of $n = n$

Level 2 of Tree: 2 Nodes of $n/2 = n$

Level 3 of Tree: 4 Node of $n/4 = n \dots$

- Stop subdividing when there are two elements left.

$n/2^k < 2$, implying $n = 2^{(k+1)}$ or $k = \log_2(n) - 1$

$T(n) = [n + 2n/2 + 2^2n/2^2 + 2^kn/2^k]$

$n[1 + 2/2 + 2^2/2^2 + 2^k/2^k]$

$n[1 + 1 + 1 + \dots k]$

$n[((2)^{(k+1)} - 1) / 2 - 1]$

$nk \mid k = \log_2(n) - 1$

$(n)(\log_2(n) - 1)$

Therefore, the upper bound runtime of the relation is $O(n \log n)$

3C.

$C = 1$, so doesn't really factor in.

Level 1 of Tree: 1 Node of $n = n$

Level 2 of Tree: 1 Nodes of $n/3 = n/3$

Level 3 of Tree: 1 Node of $n/9 = n/9 \dots$

$n/3^k < 3$, implying $n = 3^{(k+1)}$ or $k = \log_3(n) - 1$

$T(n) = [n + n/3 + n/3^2 + n/3^k]$

$n[1 + 1/3 + 1/9 + 1/3^k]$

- Geometric series?

$n(1 - 1/3^{k+1} / 1 - 1/3)$

- As k gets larger, the exponent number goes to 0.

$n(1/2/3) = (3/2)n$

$3/2n = O(n)$

Therefore, the upper bound for this relation is $O(n)$

3D.

$C = 1$, so doesn't really factor in.

Level 1 of Tree: 1 Node of $n = n$

Level 2 of Tree: 9 Nodes of $n/3 = 3n$

Level 3 of Tree: 81 Node of $n/3 = 27n \dots$

$n/3^k < 3$, implying $n = 3^{(k+1)}$ or $k = \log_3(n) - 1$

$T(n) = [n + 9n/3 + 9^2n/3^2 + 9^k n/3^k]$

$n[1 + 3 + 9^2/3^2 + 9^k/3^k]$

$n[1 + 3 + 3^2 + 3^k]$

- Geometric series and $k = \log_3(n) - 1$

$n * (3^{(\log_3(n))} - 1)/2$

$n * (n-1)/2$

- Constants like $1/2$ and -1 do not matter.

Therefore the upper bound for the run time of the relation is $O(n^2)$

3E.

$$T(3) = T(1) + 3^2 = 10$$

$$T(4) = T(2) + 4^2 = 17$$

$$T(5) = T(3) + 5^2 = 35$$

$$T(6) = T(4) + 6^2 = 53$$

$$T(7) = T(5) + 7^2 = 84$$

$$(i=1, k) \sum i^2 = k^3/3 + k^2/2 + k/6$$

Substitute n for k, it equates to $n^3/3 + n^2/2 + n/6$

Since the highest power is the only thing that matters, the upper bound of the run time of the relation is $O(n^3)$

4A.

$$A = 4, B = 2, f(n) = n$$

$$n^{(\log_b(a))} = n^{\log_2(4)} = n^2$$

$f(n) < n^2$, therefore the first case of the masters theorem should be used.

For some constant value ϵ where $1 \leq \epsilon < 2$, the bounds of the recurrence relation is found to be

$$O(n^{(\log_2(4) - \epsilon)}) \rightarrow O(n^2)$$

4B.

$$A = 2, B = 2, f(n) = n \log n$$

$$n^{(\log_b(a))} = n^{\log_2(2)} = n^1$$

$f(n) > n^1$, therefore the third case of the masters theorem should be used.

Since $f(n) = \Omega(n^{\log_b a + \epsilon})$ for some constant $\epsilon > 0$, and if $2 * n \log n / 2 \leq c * n \log n$ for some constant $c < 1$, then $T(n) = O(n \log n)$

Therefore, the upper bound of the runtime of the relation is **$O(n \log n)$**

4C.

$$A = 3, B = 3, f(n) = n$$

$$n^{(\log_b(a))} = n^{\log_3(3)} = n^1$$

$f(n) = n^1$, therefore the second case of the masters theorem should be used.

$f(n)$ and $O(n \log_b a)$ are equal with the constant of 1, so the upper bound of the run time of recurrence relation $T(n) = O(n \log n)$

4D.

$$A = 2, B = 4, f(n) = \text{root}(n) \text{ or } n^{1/2}$$

$$n^{(\log_b(a))} = n^{\log_4(2)} = n^{1/2}$$

$f(n) = n^{1/2}$, therefore the second case of the masters theorem should be used.

$f(n)$ and $(n \log_b a)$ are equal within the constant of 1, so the upper bound of run time of recurrence relation $T(n) = O(n^{1/2} \log n)$

5A.

$f(n)$ = If $A.length = 1$, then increments first element by 1. This is a base case and a constant operation with runtime $O(1)$.

If $A.length > 1$, the array is split in half and two calls made on array size of $n/2$.

Third call is made with $n = 1$.

C = constant time ops.

So $T(n) = 2T(n/2) + T(1) + c$

- $T(1)$ is also a constant operation.

$T(n) = 2T(n/2) + c$

Therefore the recurrence relation is:

$T(n) = 2T(n/2) + c$ (when $n > 1$)

1 (c) (when $n = 1$)

5B.

$T(n) = 2T(n/2) + c$

$A = 2, B = 2, f(n) = c$

$n^{(\log_b(a))} = n^{\log_2(2)} = n^1$

$f(n) < n^1$

This is an instance where case 1 of the master's theorem is applied, as $f(n)$ is smaller in runtime than $n^{\log_b a}$.

Therefore for constant $0 \leq \epsilon < 1$, the upper bound of the recurrence relation runtime is

$T(n) = O(n)$